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Nocturnal activity of phlebotomine sandflies (Diptera: Psychodidae) in a cutaneous leishmaniasis focus in Chichaoua, Morocco

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Abstract The nocturnal activity of phlebotomine sandflies (Diptera: Psychodidae) was studied “at an epidemic focus” on human cutaneous leishmaniasis due to *Leishmania tropica* Wright in Chichaoua province, in Morocco. Sandflies were collected using light and sticky-paper traps changed at 2-h intervals, inside and around houses, in August and October 2004. Overall, 633 sandflies, belonging to six species of *Phlebotomus* and three of *Sergentomyia*, were collected. Sandfly activity was nocturnal and higher at twilight. Several activity patterns were observed according to the species. *Phlebotomus* (*Paraphlebotomus*) *sergenti* Parrot, 1917, the suspected vector of *L. tropica* in this focus, was caught during each collection performed from 1900 to 0500 hours, the numbers of species caught peaked at 1900–2100 hours. There were seasonal variations of the nocturnal activity, which could be related to the variations in temperature and relative humidity.

Introduction

Phlebotomine sandflies (Diptera: Psychodidae, Phlebotominae) play an important role in human diseases, mainly as vectors of *Leishmania* parasites. In Morocco, leishmaniasis

is an endemic disease. In the last years, the reported cases, specially that of cutaneous leishmaniasis (CL), have increased, and they were recorded in areas that previously were known to be non-endemic by the public health services.

The province of Chichaoua is an emerging epidemic focus of CL due to *Leishmania tropica* Wright [Moroccan Ministry of Public Health (MPH) 2001]. In this region, visceral leishmaniasis (VL) is sporadic, while human CL poses an important public health problem. Thus, 1,025 cases were officially reported between 2000 and 2002.

Current investigations (Guernaoui et al. 2005a) had studied the sandfly fauna in this focus and had suspected *Phlebotomus* (*Paraphlebotomus*) *sergenti* Parrot, 1917 to be the vector responsible for *L. tropica* transmission. Ten species of sandflies were recorded, and they showed a marked seasonality in relation to their bioclimatic preferences. Furthermore, more studies are needed to provide a good knowledge of leishmaniasis transmission dynamics.

This paper gives the results of a study on the nocturnal activity patterns of *P. sergenti*, *Phlebotomus* (*Larroussius*) *pernicius* Newstead, 1911, *Phlebotomus* (*L.*) *ariasi* Tonnoir, 1921 and other sandflies, and the effect of the local weather (temperature, relative humidity) on their activity in the epidemic CL focus in Chichaoua province in Morocco.

Materials and methods

Study site

The study was carried out in one village, Lalla Aziza (31°4' N, 8°41'W, 1148 m a.s.l.), in the province of Chichaoua in Morocco.

This site was selected on two criteria. The first one was the epidemiological interest related to the occurrence of CL due to *L. tropica* (MPH 2001; Guernaoui et al. 2005a). The second one was the specific richness and abundance of sandflies in this site (Guernaoui, unpublished data).

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Table 1 Numbers of male (M) and female (F) of sandflies collected with sticky and light traps in entomological surveys in Lalla Aziza village (Morocco), in August and October 2004

Species	August					October				
	Sticky trap		Light trap		Total	Sticky trap		Light trap		Total
	M	F	M	F		M	F	M	F	
<i>Phlebotomus papatasi</i>	1	2	4	5	12	3	0	0	2	5
<i>Phlebotomus sergenti</i>	17	9	18	34	78	2	2	0	5	9
<i>Phlebotomus alexandri</i>	0	0	0	0	0	0	2	0	1	3
<i>Phlebotomus longicuspis</i>	11	2	1	3	17	2	0	1	1	4
<i>Phlebotomus perniciosus</i>	10	1	2	3	16	209	68	46	71	394
<i>Phlebotomus ariasi</i>	2	1	0	2	5	2	0	10	20	32
<i>Sergentomyia fallax</i>	11	5	1	1	18	5	3	1	0	9
<i>Sergentomyia minuta</i>	3	3	1	1	8	0	0	0	0	0
<i>Sergentomyia dreyfussi</i>	1	5	1	3	10	1	9	1	2	13
Total	56	28	28	52	164	224	84	59	102	469

Sandfly collection

General activity, including that of males, was studied by trapping sandflies throughout the night in August and October 2004, which correspond to periods of strong activity of *P. sergenti* and of the *Larroussuis* subgenus (mainly *P. perniciosus* and *P. ariasi*), respectively (Guernaoui et al. 2005a).

Sandflies were collected with two CDC miniature light traps and 30 sticky traps (A4 white paper coated with castor oil). Traps were placed before sunset, changed at 2-h intervals, until there was no active sandfly (after sunrise).

Collections were made inside and around houses. In the domestic sites, traps were placed in the bathroom, kitchen, bedroom, windows and in the vegetation in the hall. In the peridomestic ones, they were placed close to the vegetation around human dwellings, crevices in the animal sheds, chicken coops and tree holes.

Specimens caught were preserved in 70% alcohol, cleared in potassium chlorate and mounted in Canada balsam. Identification was based on the morphology of the male genitalia and of female spermathecae and pharynges. The differentiation between males of *P. perniciosus* and that of *P. (L.) longicuspis* Nitzulescu, 1930 was made by

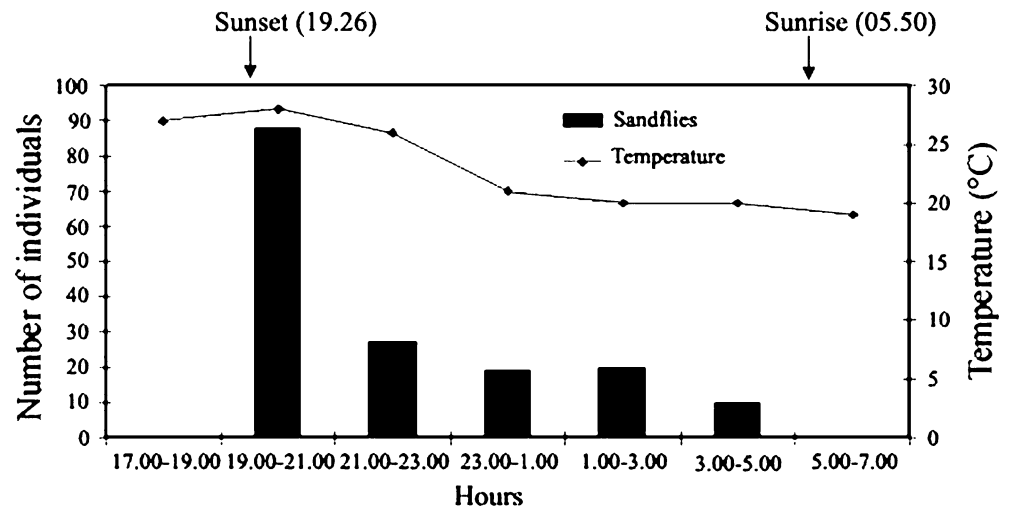
examining both the form of copulatory valves and the number of coxite hairs (Pesson et al. 2004; Guernaoui et al. 2005b).

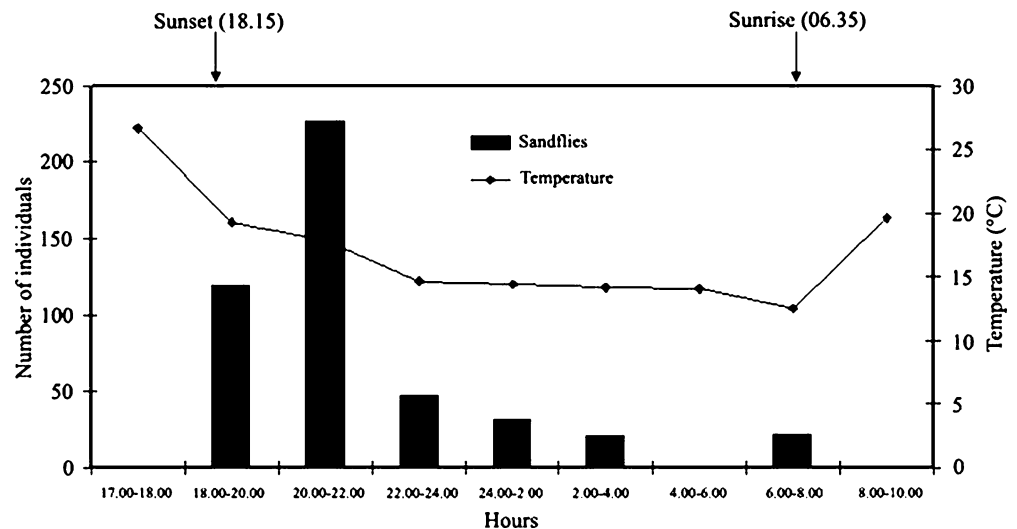
Recording climatic factors in the study site

To study the effect of weather on sandfly activity, we recorded, at 2-h intervals, the temperature and relative humidity in the study site. During all the trapping period, wind velocity (recorded using an anemometer) was constant and equal to zero; hence, we could not evaluate its impact on the sandfly activity in this study.

Results

Overall, 633 phlebotomine sandflies, belonging to six species of *Phlebotomus* and three of *Sergentomyia*, were collected (Table 1). The sandfly diversity was more significant in August (relative Shannon index=0.8) due, mainly, to the prevalence of *P. perniciosus* in October (relative Shannon

Fig. 1 Nocturnal activity of the sandfly fauna in August 2004 in Lalla Aziza. 109×58 mm (200×200 DPI)



index=0.3). The species prevalence varied with the collecting period.

Concerning species of *Phlebotomus*, *P. sergenti* (47.6%) predominated in August, followed by *P. longicuspis* (10.4%), *P. perniciosus* (9.8%) and *P. ariasi* (3%). *P. (Paraphlebotomus) alexandri* Sinton, 1928 was absent. In October, *P. perniciosus* was the predominant species (84%), followed by *P. ariasi* (6.8%). The other *Phlebotomus* species were not commonly observed.

Sergentomyia species were more prevalent in August. *Sergentomyia* (*Sergentomyia*) *minuta* Rondani 1843 was absent in October.

The data analysis (Figs. 1 and 2) shows that the sandfly activity was crepuscular and nocturnal. No phlebotomine sandfly was collected before sunset or after sunrise. The activity period of the total fauna varied with the photoperiod. In August (summer), when the duration of day was longer, sandflies were active during 10 h from 1900 to 0500 hours, whereas in October (autumn), they were active during 14 h from 1800 to 0800 hours.

For both seasons, the activity occurred throughout the night and was highest at twilight: 70 and 73% of sandflies were collected within 2 h of sunset in August and October, respectively.

Several activity patterns were observed according to the species and the season. In August (Fig. 3), the species *P. sergenti*, *P. perniciosus* and *S. minuta* were active all night with a highly marked crepuscular activity. On the other hand, *P. longicuspis*, slightly active at twilight, displayed a nocturnal and constant activity, while the activity of *P. ariasi* started at midnight. *Sergentomyia* (*Grassomyia*) *dreyfussi* Parrot, 1933 displayed a crepuscular activity. It was collected only between 1900 and 2100 hours and between 0300 and 0500 hours.

In October (Fig. 4), sandfly activity was strong until midnight. During this part of the night, *Sergentomyia* (*S.*) *fallax* Parrot, 1921 reached a peak between 1800 and 2000 hours, *P. (Phlebotomus) papatasi* Scopoli, 1786 and *P. sergenti* between 2000 and 2200 hours, and *P. alexandri* was collected only between 2200 and 2400 hours.

After midnight, only the *Larroussuis* species (*P. longicuspis*, *P. perniciosus* and *P. ariasi*) and *S. dreyfussi* were collected. A break in activity was also observed between 0400 and 0600 hours. *P. perniciosus* and *S. dreyfussi* were active throughout the night with a peak at twilight. *P. ariasi* reached a peak between 2200 and 2400 hours, while *P. longicuspis* was active only at midnight.

Fig. 3 Hourly activity of sandfly species in August in Lalla Aziza. *Number multiplied by 10^{-1} . 109×54 mm (200×200 DPI)

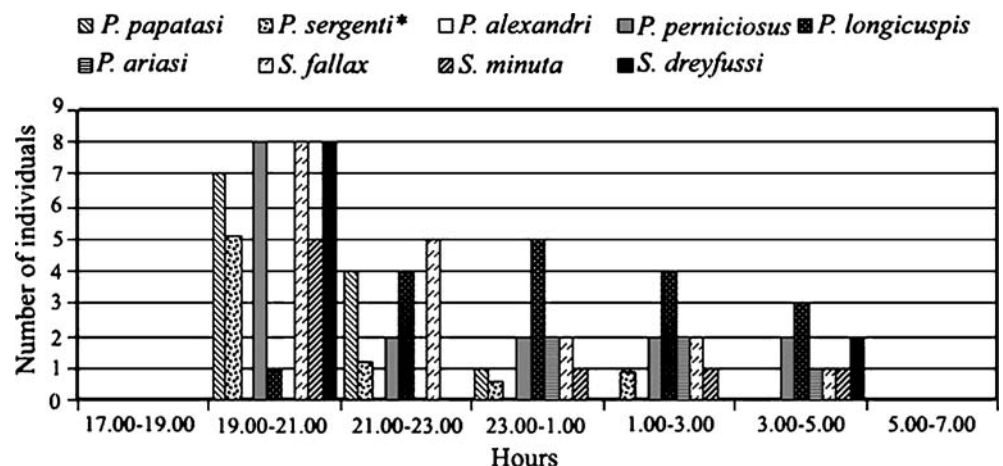
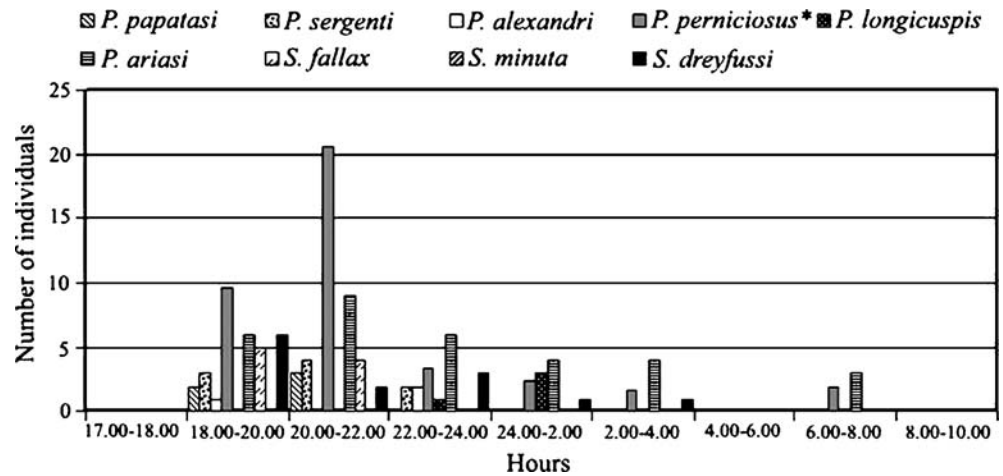


Fig. 4 Hourly activity of sandfly species in October in Lalla Aziza. *Number multiplied by 10^{-1} . 109×52 mm (200×200 DPI)



The effect of weather on sandfly activity

In this study, we highlighted the existence of a correlation between the activity of sandflies, temperature and relative humidity in this station. The variation in temperature and relative humidity during the collecting nights is shown in Figs. 5 and 6.

The species *P. sergenti*, *P. perniciosus*, *S. fallax* and *S. dreyfussi* showed a significant correlation, negative with the relative humidity [$r=-0.9$ ($p=0.01$), $r=-0.9$ ($p=0.01$), $r=-0.9$ ($p=0.01$), $r=-0.7$ ($p=0.05$), respectively] and positive with temperature [$r=0.8$ ($p=0.05$), $r=0.8$ ($p=0.01$), $r=0.9$ ($p=0.01$), $r=0.9$ ($p=0.05$), respectively]. *P. papatasi* showed a significant negative correlation with the relative humidity ($r=-0.9$, $p=0.01$), while *P. ariasi* showed a significant negative correlation with temperature ($r=-0.9$, $p=0.01$).

Discussion

The study of the sandfly nocturnal activity is significant in that it reveals the best time to collect and study the flies. It is also indicative of the time when a person is mostly likely to be bitten and to get leishmaniasis.

The nocturnal activity of eight sandfly species was studied. The activity was nocturnal and stronger at twilight. It started at sunset and stopped immediately after sunrise. Its duration was directly dependent on the photoperiod. This nocturnal rhythm could result from the decline in the diurnal temperature and the increase in relative humidity (Cross and Hyams 1996).

Sandfly activity is known to be affected by several environmental conditions, such as temperature, humidity, rainfall and wind velocity (World Health Organization 1990). In this study, the activity varied with the species and the season.

In August (summer), sandfly activity was continuous all night because the climatic factors, especially temperature and relative humidity, were favourable. In this period, *P. sergenti*, the proven vector of *L. tropica* in Morocco (Guilvard et al. 1991), was caught during each collection from 1900 to 0500 hours, the numbers of species caught peaked at 1900–2100 hours. However, there were seasonal variations, being slight at the time of this peak and strong in the activity duration.

P. perniciosus and *P. ariasi*, the main vectors of *L. infantum* Nicolle, 1908 in the Mediterranean basin (Rioux et al. 1986), were characterized by their seasonal peaks in autumn (Guernaoui et al. 2005a). During this period

Fig. 5 Hourly variation in temperature and relative humidity in August in Lalla Aziza. 109×53 mm (200×200 DPI)

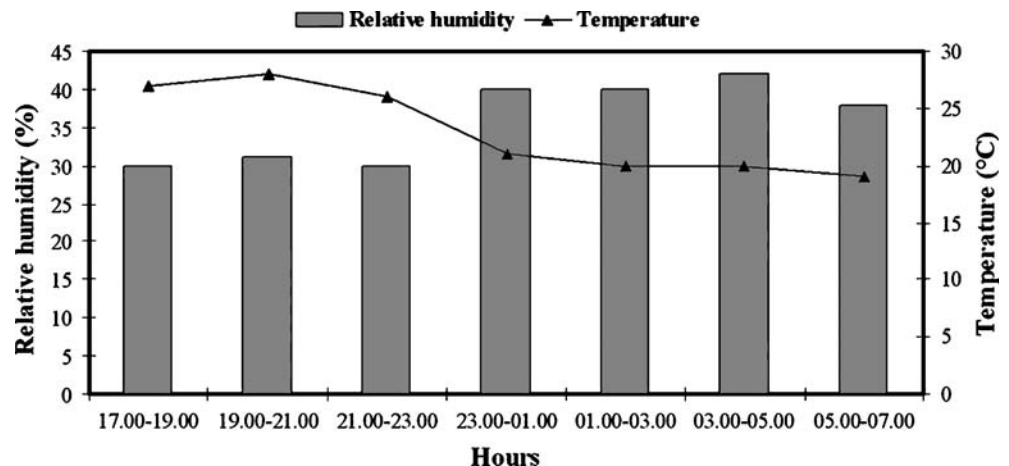
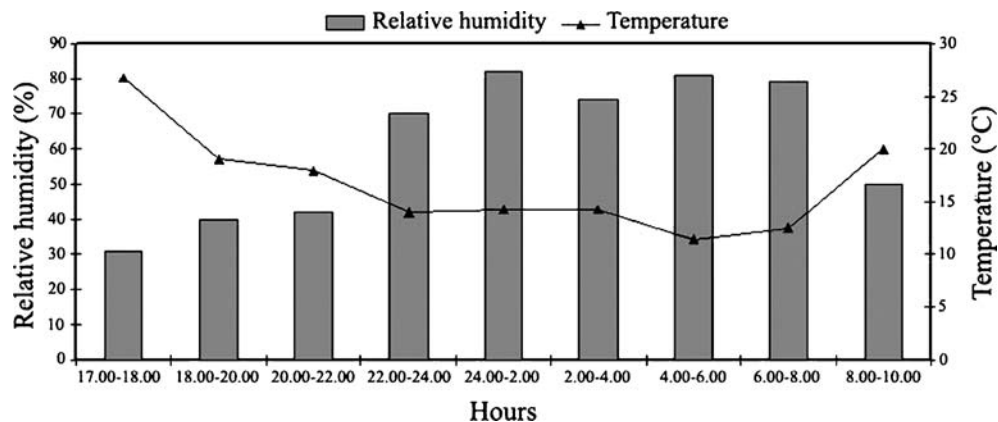


Fig. 6 Hourly variation in temperature and relative humidity in October in Lalla Aziza.
109×48 mm (200×200 DPI)



(October), sandfly activity declined significantly after midnight. *P. perniciosus* was active all night and peaked at 20.00–22.00 hours. The activity of *P. ariasi* started immediately after sunset and increased afterwards, reaching a maximum around midnight when temperature tends to be lower and humidity higher. The same pattern was observed for *P. papatasi* in Central Iraq (Mohsen 1983).

In this study, we demonstrated the existence of seasonal variations in the nocturnal activity. There were substantial differences in the distribution of the nocturnal activity related to the intensity and the activity peaks. Dinesh et al. (2001) wondered why the time of the night when landing/biting by *P. argentipes* peaks should vary with season. They added that it may partly reflect seasonal changes in the sleeping patterns of the villagers. We demonstrated here that such differences should be, principally, correlated with changes in climatic factors. The temperature and relative humidity were determining factors of sandfly activity. *P. sergenti* and *P. perniciosus* ‘preferred’ low humidity (40–50%) and high temperature (>20°C), which could explain that their activity was stronger at twilight. In contrast, in summer, *P. ariasi* activity started only at midnight when humidity was high (70–80%) and temperature was low (15°C). Different hourly patterns per se are significant. They allow the species to adapt to the climatic variations and could be used to reduce competition between sympatric species.

In conclusion, in this focus, in addition to the marked seasonality, phlebotomine sandflies also showed different nocturnal activity patterns which are managed by climatic factors. *P. sergenti* peak transmission of *L. tropica* to humans probably occurs at twilight.

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